AI SOCIAL ENGINEERING
THREAT INTELLIGENCE REPORT

Artificial Intelligence Cyber Threat Intelligence

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EXECUTIVE SUMMARY

AI Social Engineering is a growing cybersecurity threat where attackers use artificial intelligence to trick people into taking harmful actions [1]. Instead of breaking into systems using technical methods, the attacker focuses on manipulating human trust[1], [2].

In this type of attack, tools like voice cloning and automated messaging are used to impersonate trusted individuals such as company executives [1], [2]. A real example of this involved a company employee receiving a phone call that sounded exactly like their CEO. Believing the request was genuine, the employee transferred a large amount of money to the attacker [2].

This attack mainly affects organizations where employees handle financial transactions or sensitive information [1]. It is especially dangerous because it does not rely on malware or system vulnerabilities, making it harder to detect using traditional security tools [1], [2].

Understanding how this attack works is important for preventing financial loss, protecting sensitive data, and strengthening overall organizational security.

ATTACK OVERVIEW

AI Social Engineering attacks rely on the manipulation of human behaviour rather than technical system vulnerabilities. In this attack, the attacker uses artificial intelligence tools, such as voice cloning or automated messaging systems, to impersonate a trusted individual within an organization [1], [2].

The attack begins with reconnaissance, where the attacker gathers information about the target organization, including employee roles, communication patterns, and key decision-makers [1]. This information is often obtained from public sources such as LinkedIn, company websites, or previous data leaks [1].

Once sufficient information is collected, the attacker uses AI tools to create a convincing impersonation. In the case of voice cloning, AI is used to generate speech that closely matches the voice of a senior executive. The attacker then contacts the victim, typically through a phone call, and creates a sense of urgency to pressure the victim into acting quickly [2].

In a documented real-world case, attackers successfully used AI-generated voice technology to impersonate a CEO and instruct an employee to transfer approximately $243,000 to a fraudulent account [1], [2]. The employee, believing the request was legitimate, completed the transaction without further verification [1].

This attack highlights how AI enhances traditional social engineering techniques by increasing realism and reducing the effort required to deceive victims. Unlike traditional phishing attacks, there may be no suspicious links or malware involved, making detection more challenging [1], [2].

The $25 million video call that wasn't

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In February 2024 reports emerged of an AI-powered social engineering attack on a finance worker at multinational Arup. They'd attended an online meeting with who they thought was their CFO and other colleagues.

During the videocall, the finance worker was asked to make a $25 million transfer. Believing that the request was coming from the actual CFO, the worker followed instructions and completed the transaction.

Initially, they'd reportedly received the meeting invite by email, which made them suspicious of being the target of a phishing attack. However, after seeing what appeared to be the CFO and colleagues in person, trust was restored.

The only problem was that the worker was the only genuine person present. Every other attendee was digitally created using deepfake technology, with the money going to the fraudsters' account.

Figure 1: News report discussing an AI-powered deepfake scam incident

Table 1: ***OSINT Research and Source – AI Social Engineering Incident***

Source Name	Author/Org	Key Information Provided	Year	Source Link
The Hacker News	Ravie Lakshmanan	Detailed the $243k loss and the CEO impersonation Reported a $25 million deepfake video-call fraud [1].	2025	https://thehackernews.com/2025/01/top-5-ai-powered-social-engineering.html
The Hacker News	Ravie Lakshmanan	Explains how AI-powered chatbots on Facebook are used to harvest credentials by creating a false sense of urgency [1].	2025	https://thehackernews.com/2025/01/top-5-ai-powered-social-engineering.html
The "Deepfake CEO" Scam	Christo	Explains how AI voice cloning is replacing text-based BEC by exploiting organizational hierarchies and human trust, recommending "zero trust" voice policies and challenge-response "safe words" to prevent financial fraud [2].	2026	https://elite.nz/the-deepfake-ceo-scam-why-voice-cloning-is-the-new-business-email-compromise-bec/

CISA Spyware Warning	CISA / Cyber Security Hub	Details how threat actors use sophisticated **social engineering** and trojanized apps (masquerading as Signal/WhatsApp) to bypass encryption and hijack accounts of high-value targets [3].	2025	https://www.linkedin.com/pulse/cisa-warns-threat-actors-using-commercial-spyware-rti4e/

MITRE ATT&CK MAPPING

Figure 2: https://attack.mitre.org/techniques/T1566/004/

Table 2: ***MITRE ATT&CK Mapping – AI Social Engineering***

Tactic	Technique Name	ID	Application to Attack
Reconnaissance	Phishing for Information	T1598	Attackers gather voice samples from news interviews or social media to train AI models for impersonation. [4]
Initial Access	Spear Phishing Voice	T1566.004	Scammers use AI voice cloning and video deepfakes to trick finance workers into authorizing transfers. [5]

Initial Access	Spear Phishing Service	T1566.003	Threat actors use trojanized apps masquerading as Signal or WhatsApp to hijack high-value accounts. [6]
Persistence	Account Manipulation	T1098	Attackers exploit "linked device" functions on messaging apps to maintain long-term access to communications. [7]
Impact	Financial Theft	T1657	The goal in the documented Arup and CEO scams was the theft of $25 million and $243k respectively. [8]

Who is at Risk

AI-powered social engineering attacks pose a threat to both individuals and organizations [1]. The most at-risk groups include:

- Finance workers, executive assistants, and managers who handle payments or sensitive data [1]
- Employees targeted through executive impersonation to authorize urgent fund transfers [1][2]
- Small and medium-sized businesses in developing regions like Nigeria, where cybersecurity awareness is limited [2]

A documented case saw a finance worker lose $25 million after being deceived by a deepfake video call impersonating a CFO [1].

Detection Methods

Detecting these attacks is challenging because they are designed to appear legitimate [1]. Key indicators to watch for:

- Urgent requests involving fund transfers, credential changes, or sensitive data [1][3]
- Communication that uses pressure, fear, or urgency as a manipulation tactic [1][2]
- Inconsistencies in voice tone, video quality, or communication style [2]

Organizations should monitor unusual account activity, enforce approval processes for financial transactions, and verify sensitive requests through a secondary communication channel [1][3].

Prevention and Recommendations

Effective prevention requires a combination of employee awareness and organizational controls [1][2]. Key measures include:

- Training staff to recognize impersonation, urgency, and authority-based manipulation [1][2]

- Enforcing strict verification procedures for financial transactions and sensitive requests [2]
- Adopting challenge-response "safe words" as a countermeasure against voice cloning [2]
- Implementing multi-factor authentication and role-based access controls [3]
- Enforcing app vetting policies, as CISA warns of trojanized Signal and WhatsApp apps used to hijack accounts [3]
- Conducting regular security awareness training, phishing simulations, and incident reporting drills [1][2]

References

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